CADT បណ្ឌិត្យសភាបច្ចេកវិទ្យាឌីជីថលកម្ពុជា
Cambodia Academy of Digital Technology

IDT

Detection Consultation and Doctor Appointment

Redo Capstone 1 Defense

Advisor : Mr. Ang Mengchhuong
Major : Computer Science
Specialize : Data Science & Software Engineer

Defense Date: September 25, 2024

Agenda

1. GENERAL PRESENTATION
2. LITERATURE REVIEW
3. ANALYSIS & CONCEPTION
4. IMPLEMENTATION
5. CONCLUSION
6. DEMONSTRATION

1

GENERAL PRESENTATION

OVERVIEW



Approximately 2.2 billion people have some form of vision impairment, of which at least 1 billion cases could have been prevented.

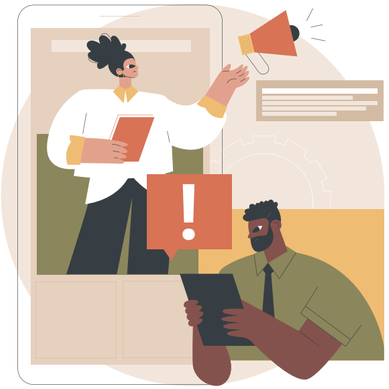
- cataract (94 million)
- glaucoma (7.7 million)
- diabetic retinopathy (3.9 million)
- refractive error (88.4 million)
- age-related macular degeneration (8 million)
- presbyopia (826 million)

WHO Global Eye Health: World report on vision. Available at:
https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment

PROBLEMATIC



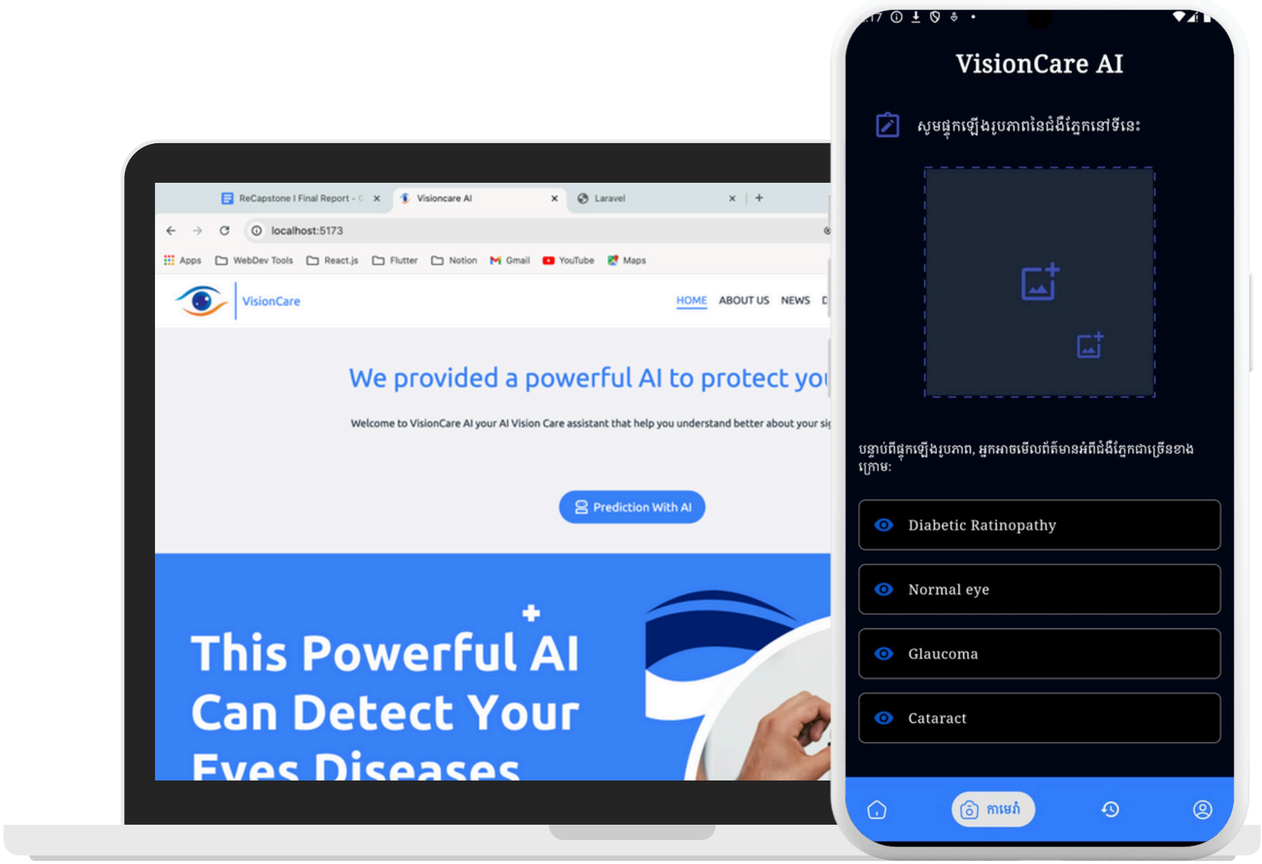
Lack of Patient Understanding and Education



Limited Access to Specialized Health Care

INTRODUCTION

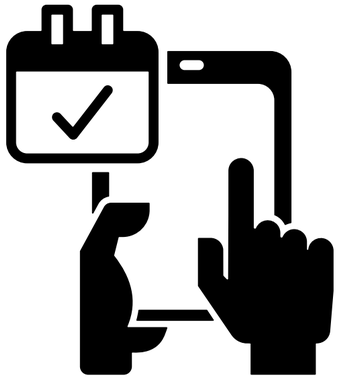
VisionCare AI is a mobile and web application that integrates with machine learning to provide a consultancy service for diagnosing eye diseases.



OBJECTIVE



Develop a platform for eye disease detection



Integrate booking system to make appointments with doctors



Provide accessible, reliable, and relevant information

TEAM STRUCTURE



Advisor
ANG Mengchhuong

Team Leader
HOUNG Sokleng





Member
RITHY Bondet



Member
HANG Rithratana



Member
EANG Kimchiv



Member
KOY Sambo



Member
HUL Sambath

TASK BREAKDOWN

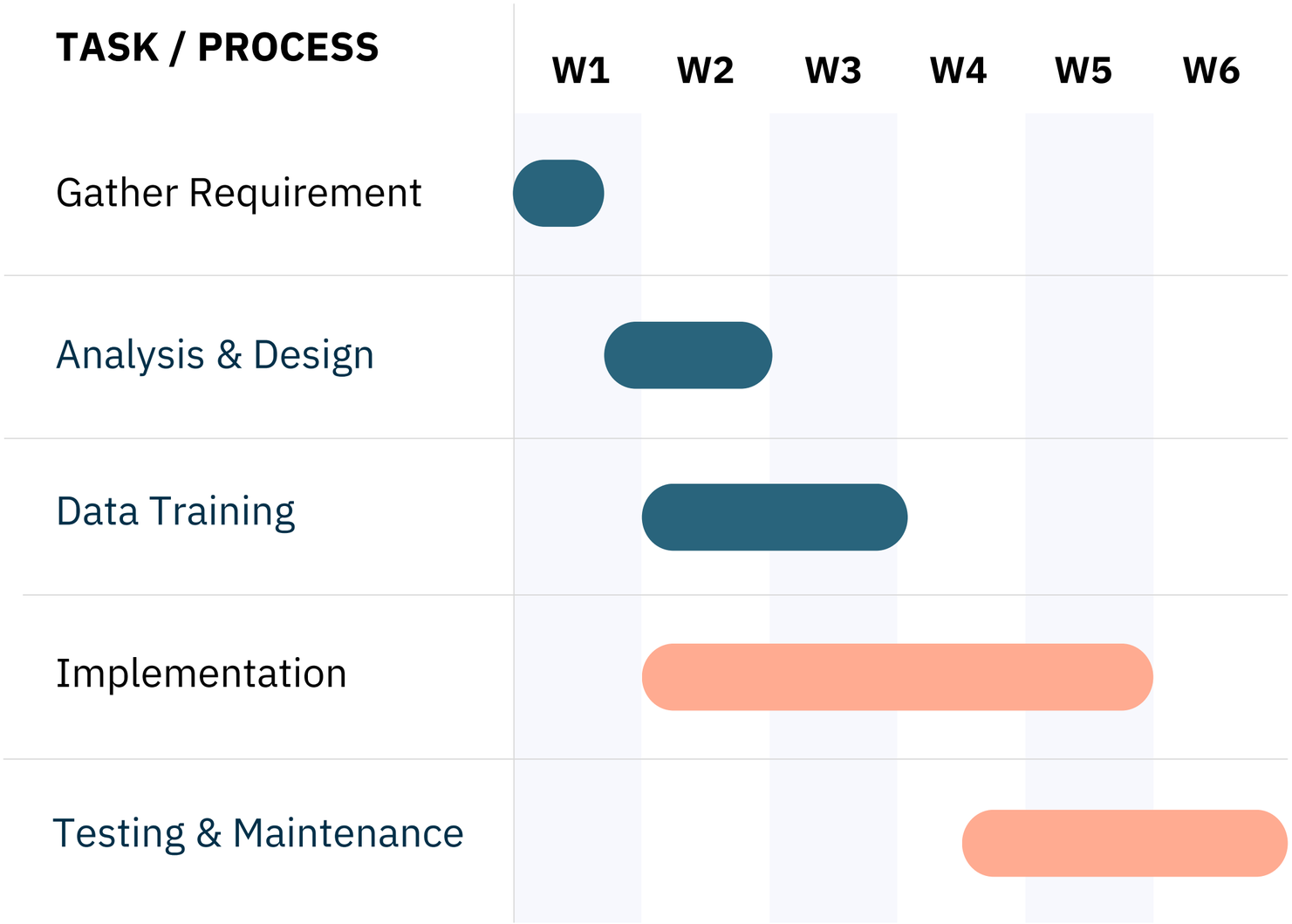
Functionalities

Mobile Application
Enhance App Theme
Offline Mode
Doctor Appointment
Website (Clinic)
View Dashboard
Manage Doctor
Respond to appointment Request

Website (User)
Authentication
User Registration
Reset Password
Edit User Profile
Dark/Light Theme
Save/Delete AI Detection History
AI Disease Detection
Send Email Directly
News Section

API
Machine Learning API Model (FLASK API)
Save Predicted History API
Clinic & Doctor API
News API
Request Appointment API
Deep Leaning
Data Collection
Data Preprocessing
Data Augmentation
DL Model Training & Evaluation

PROJECT PLANNING

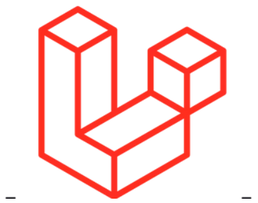




ANALYSIS & CONCEPTION

TECHNOLOGIES

Back-end & API



Laravel

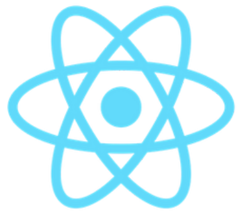


Flask

Model Training & Web Application



Python



React.js



Pytorch

Database & Mobile Application



MySQL



Flutter

TOOLS



Visual Studio Code



Draw.io



ADB



GitHub



Google Colab



Anaconda



XAMPP



Telegram



Postman



Google Meet

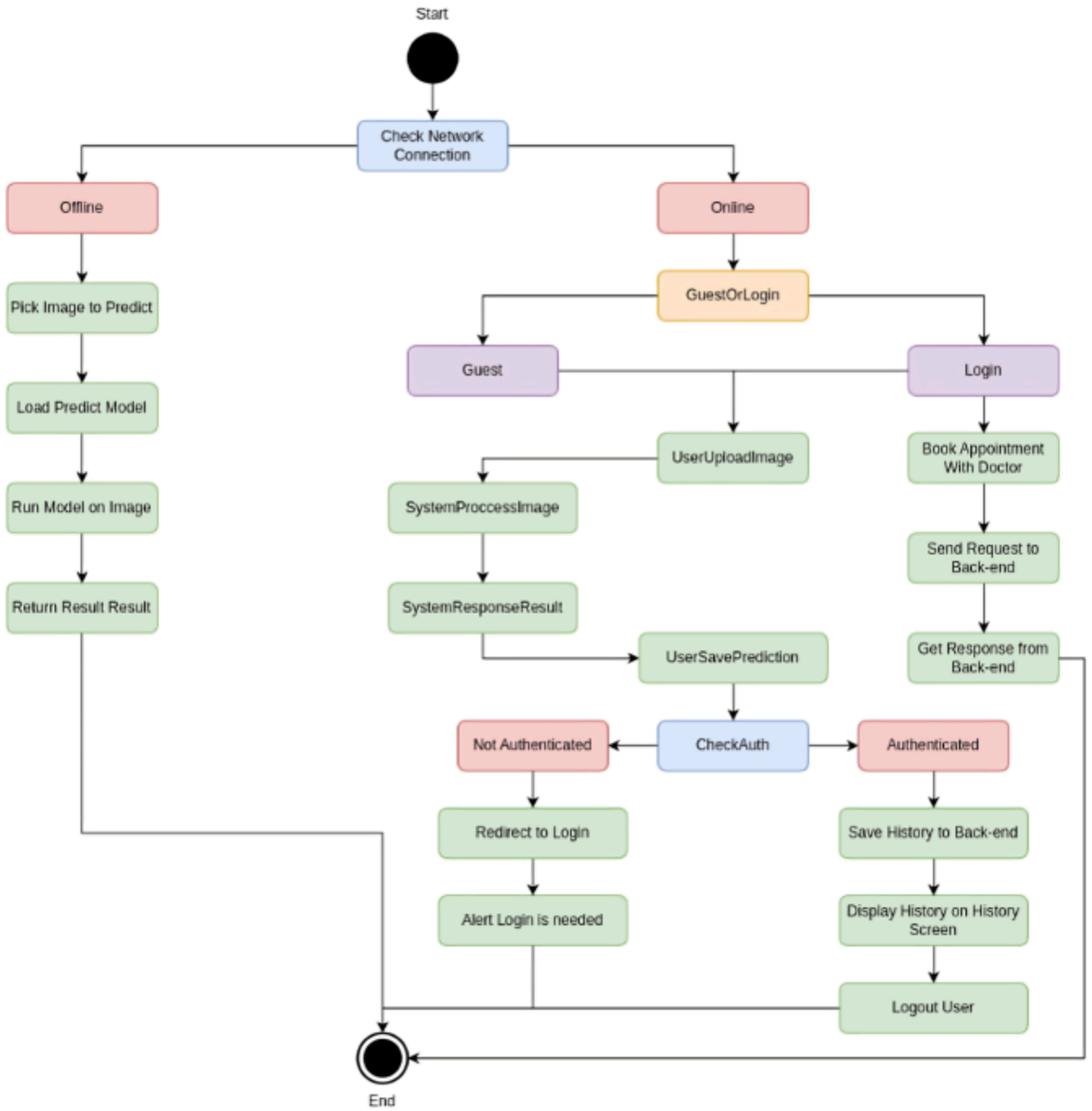
USE CASE DIAGRAM

Mobile Application



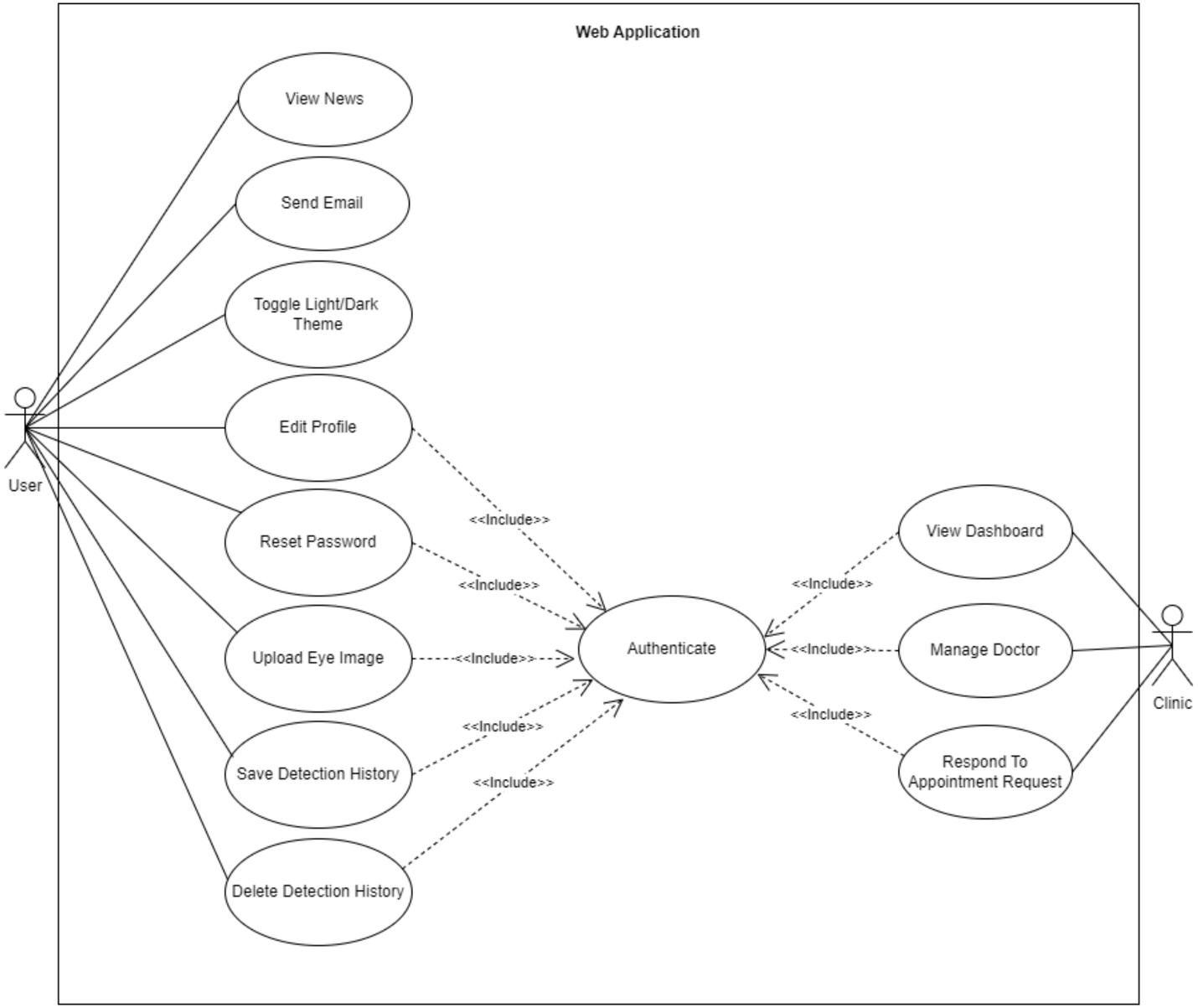
ACTIVITY DIAGRAM

Mobile Application



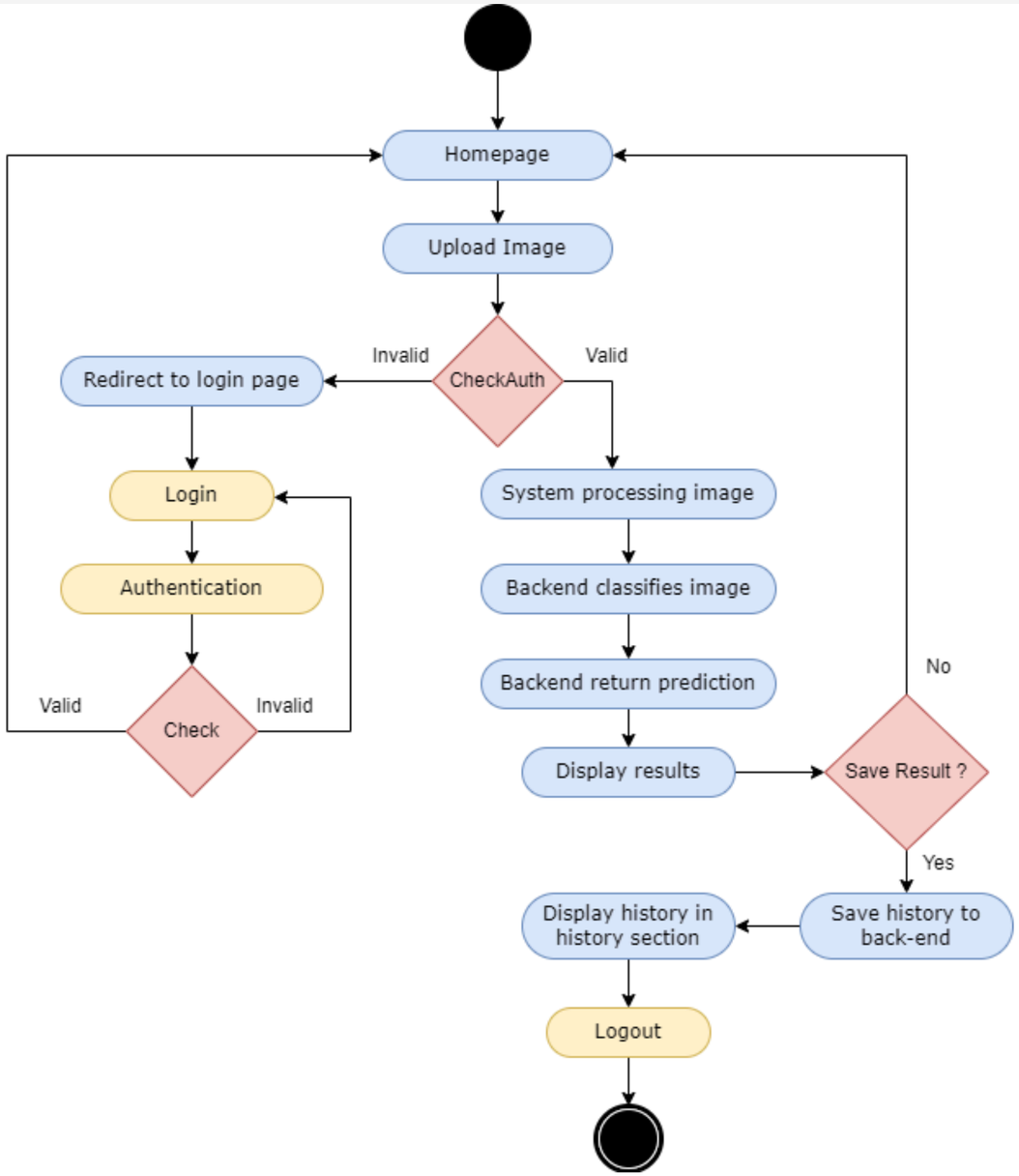
USE CASE DIAGRAM

Website

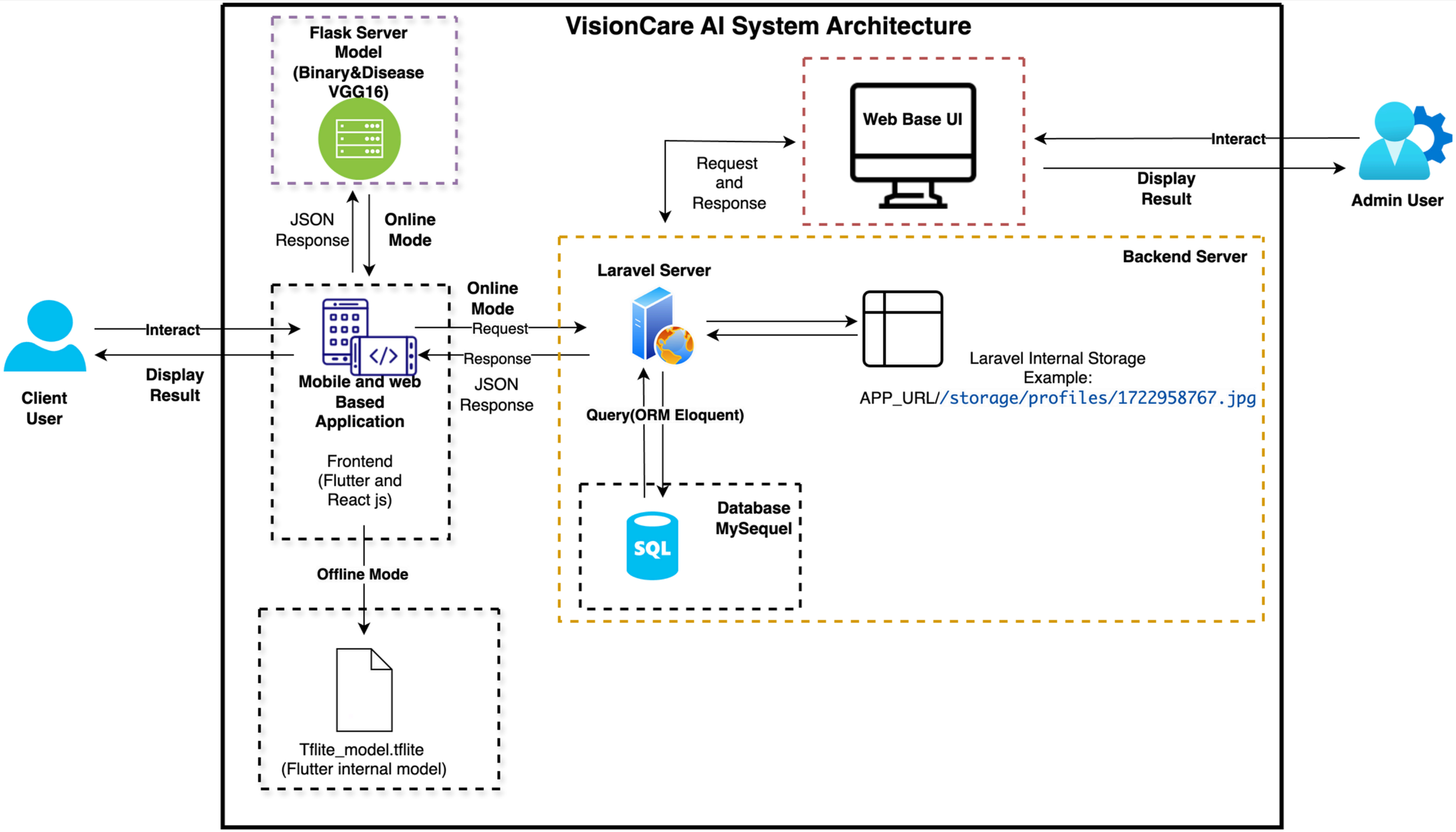


ACTIVITY DIAGRAM

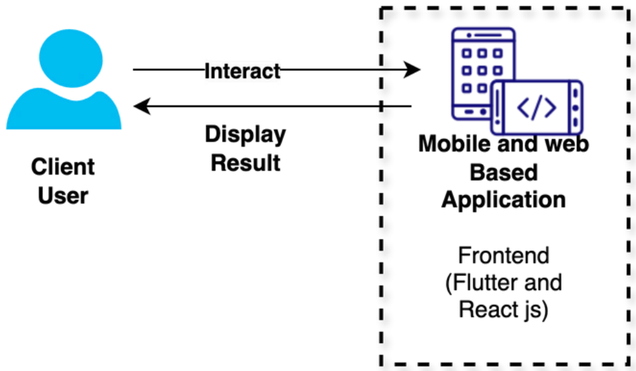
Website



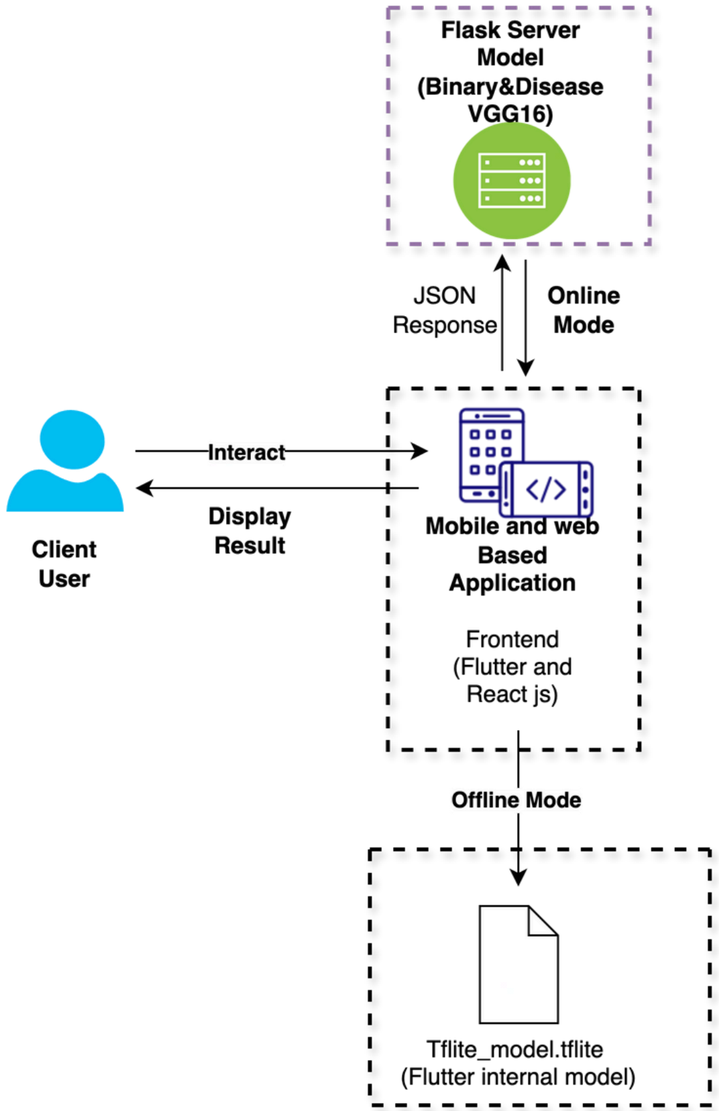
SYSTEM ARCHITECTURE



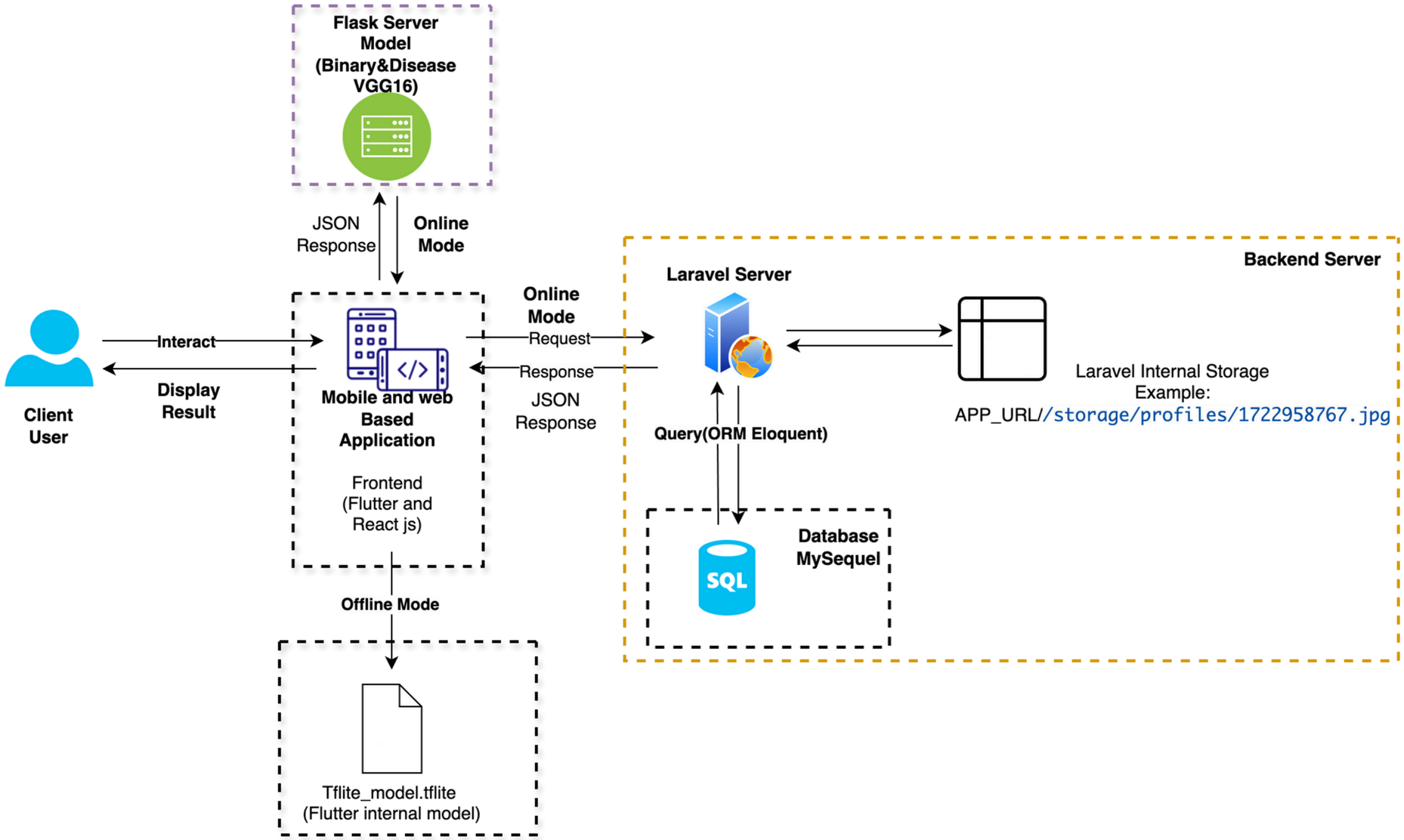
SYSTEM ARCHITECTURE



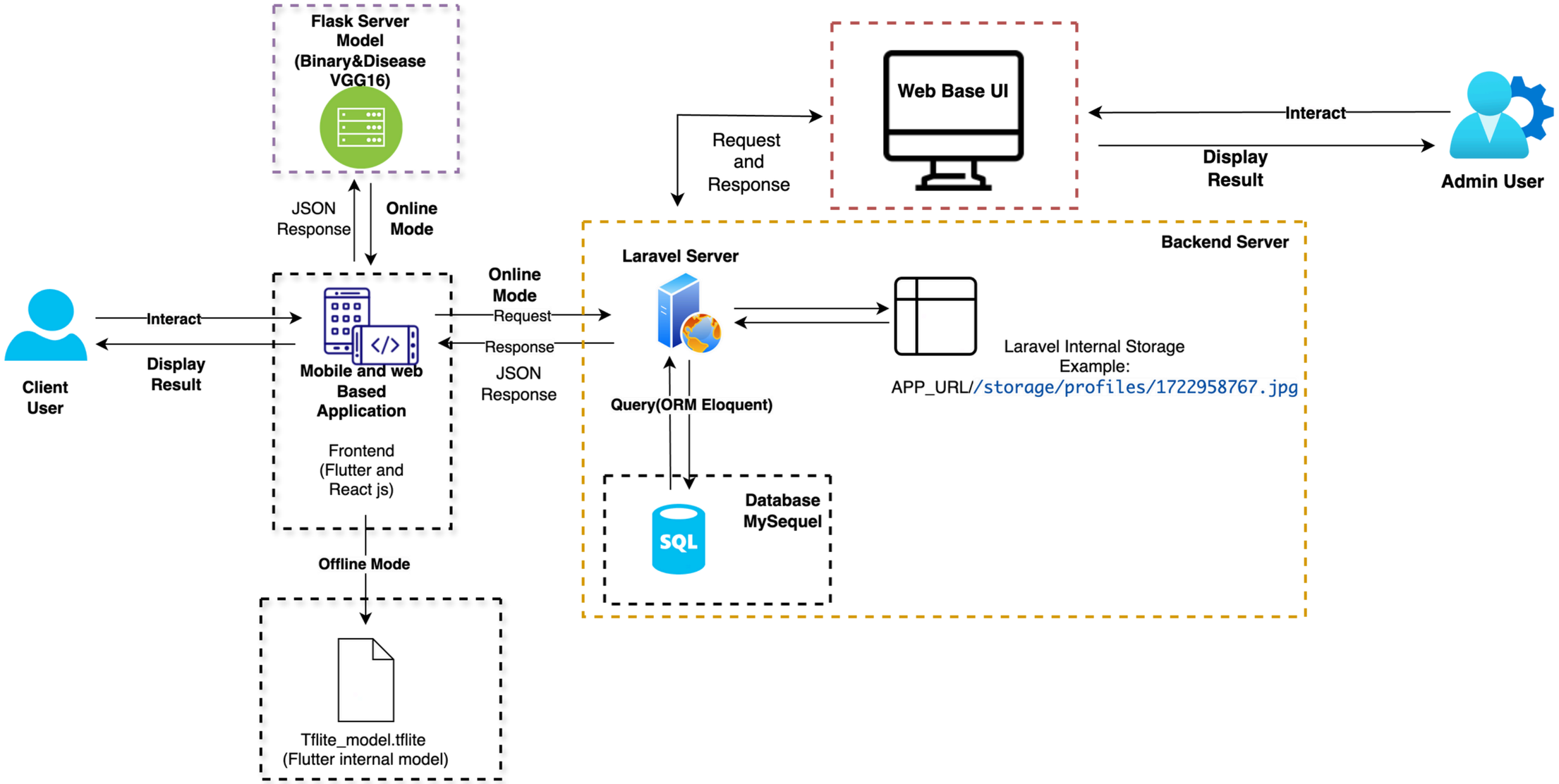
SYSTEM ARCHITECTURE



SYSTEM ARCHITECTURE



SYSTEM ARCHITECTURE



3 LITERATURE REVIEW

MODEL COMPARISION

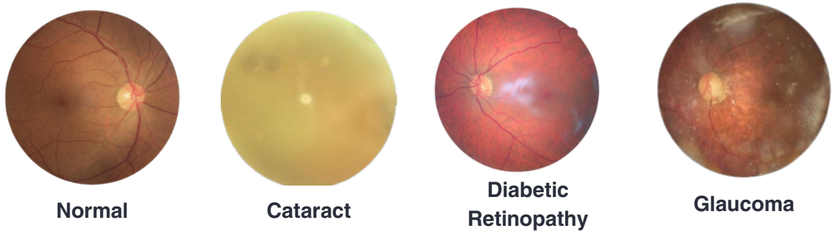
Author Name	Year	Topic	Methodology	Result	DataSet
Xu Qian	2023	Glaucoma	CNN-Based	94.53%	11,060
Ki Young Son	2022	Cataract	RestNet50	96.21%	1972
Natarajan	2024	Cataract, Glaucoma, and Diabetic-Retinopathy	VGG, RestNet, EfficientNet	94.48%	Not Included

4 IMPLEMENTATION

DATA COLLECTION

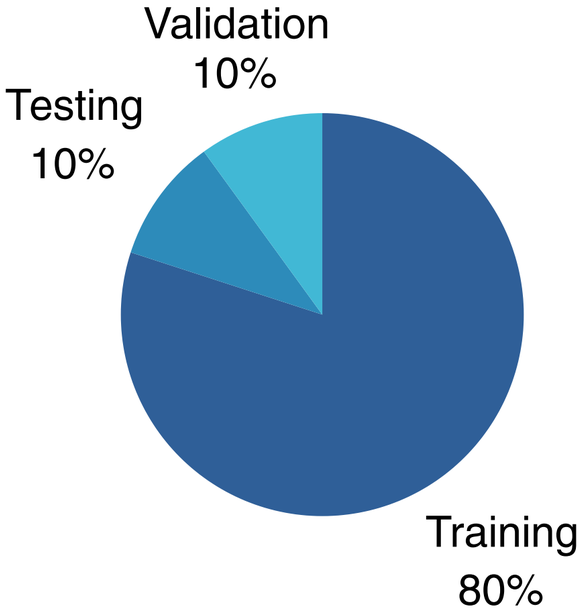
kaggle

THE LANCET Digital Health

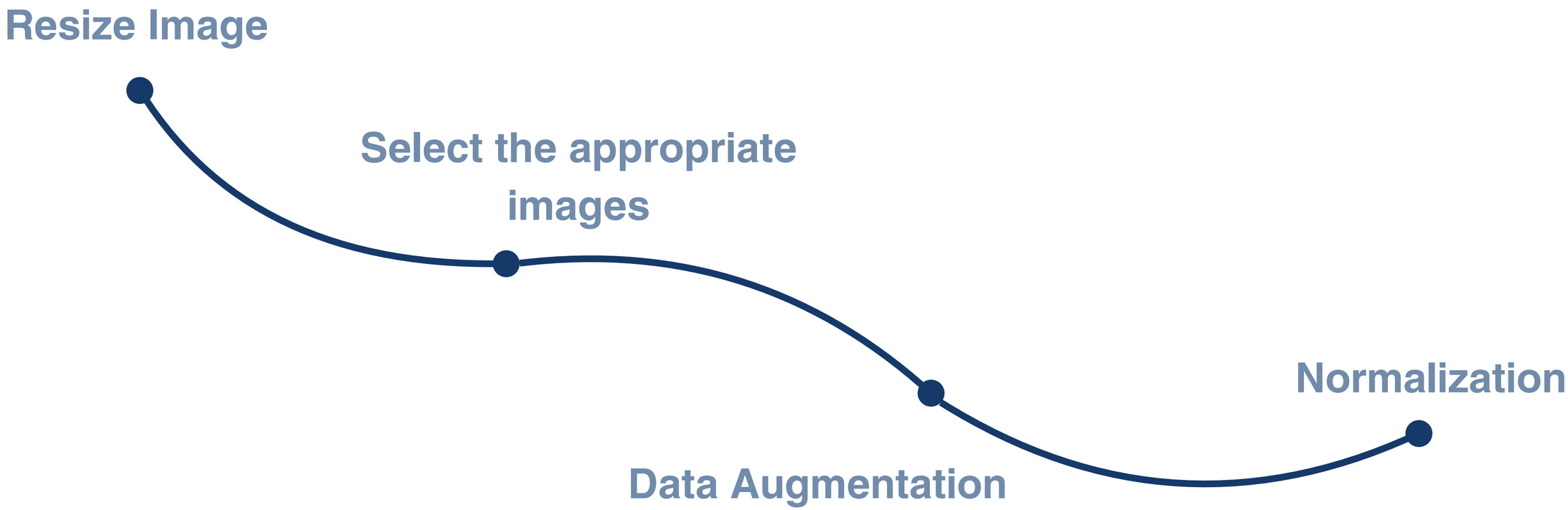


Eye	Original Data	Chose the appropriate data
Normal	520	370
Cataract	540	370
Glaucoma	538	370
Diabetic Retinopathy	526	370

After we combined the data we got around **500** images in each class. There are **4** classes.



DATA PREPROCESSING



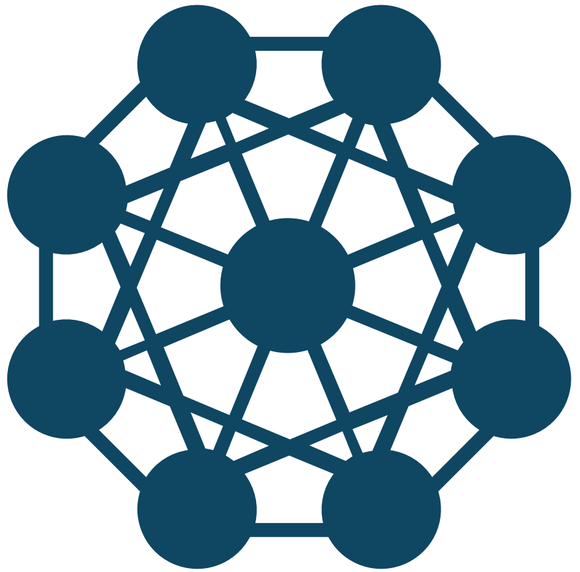
HYPER PARAMETER

Learning Rate: **0.00001**

Batch Size: **32**

Optimizer: **Adam**

Epochs: **75 (EarlyStopping)**



Model Training

MODEL SELECTION

ResNet50

VGG16

DenseNet121

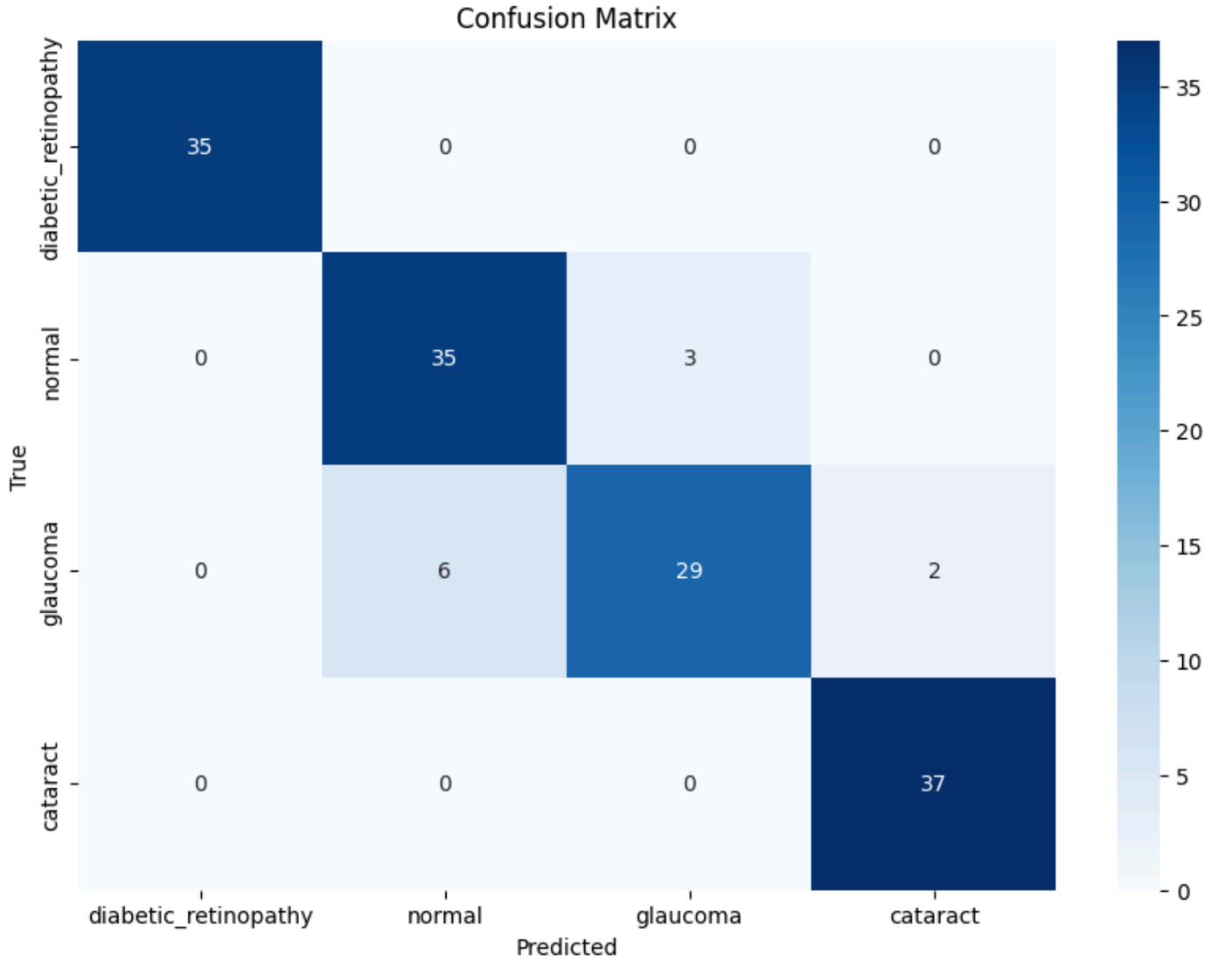
MODEL EVALUATION

Model	Learning Rate	Epoch	Accuracy (4 Classes)
Restnet	0.00001	75	71%
VGG16	0.00001	75	93%
DenseNet 121	0.00001	75	63%

VGG 16 has the highest accuracy among the other 2 models, which an accuracy was around **93%**

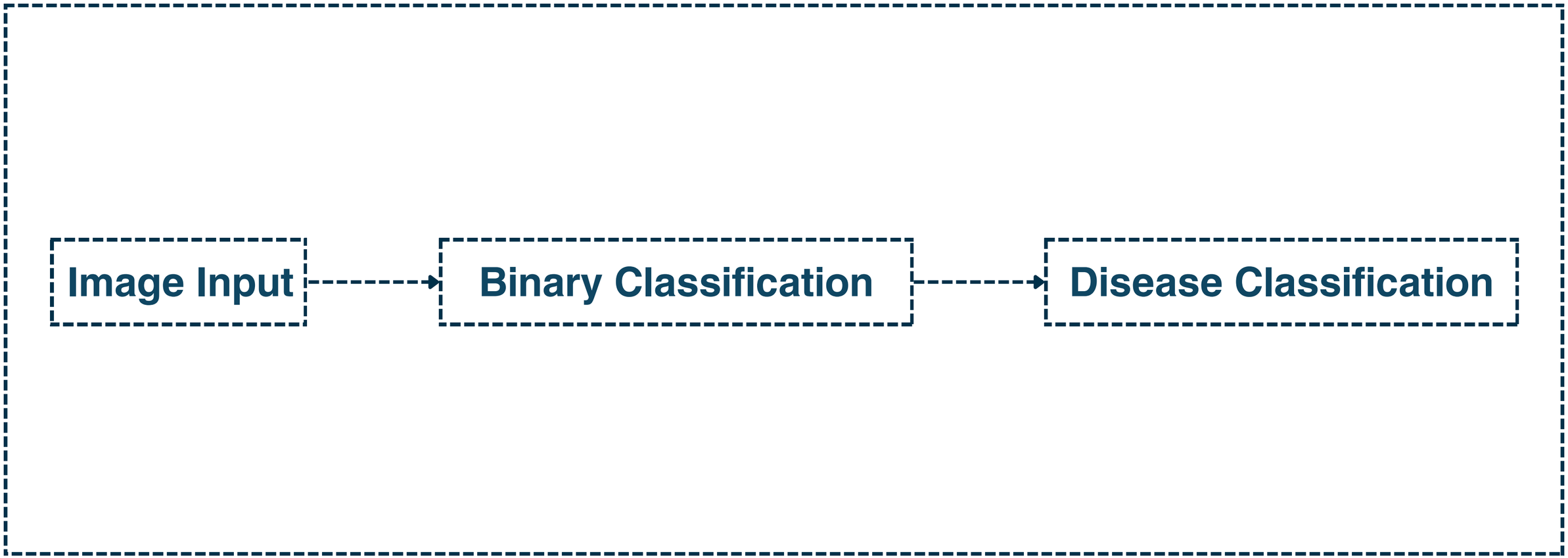
MODEL EVALUATION

Class	Precision	Recall	F1-Score
diabetic_retinopathy	1.00	1.00	1.00
normal	0.85	0.92	0.89
glaucoma	0.91	0.78	0.84
cataract	0.95	1.00	0.97



Model VGG 16

BINARY CLASSIFICATION

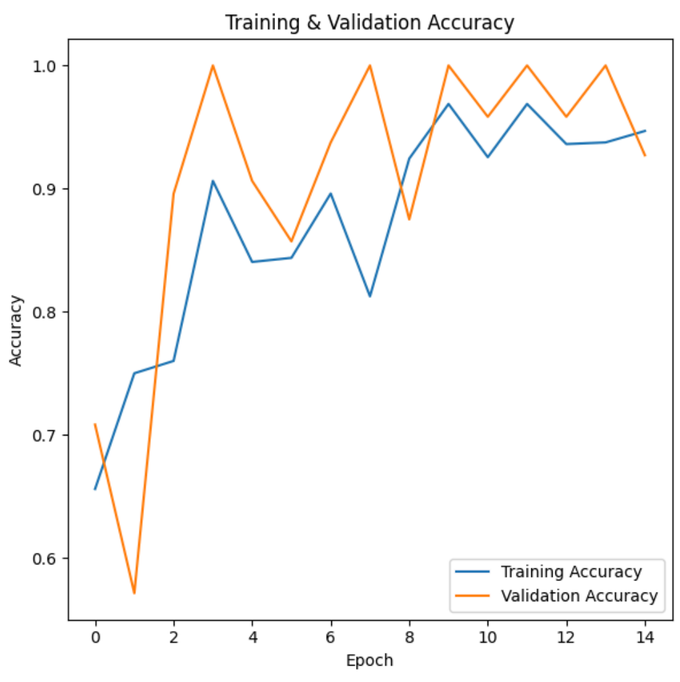


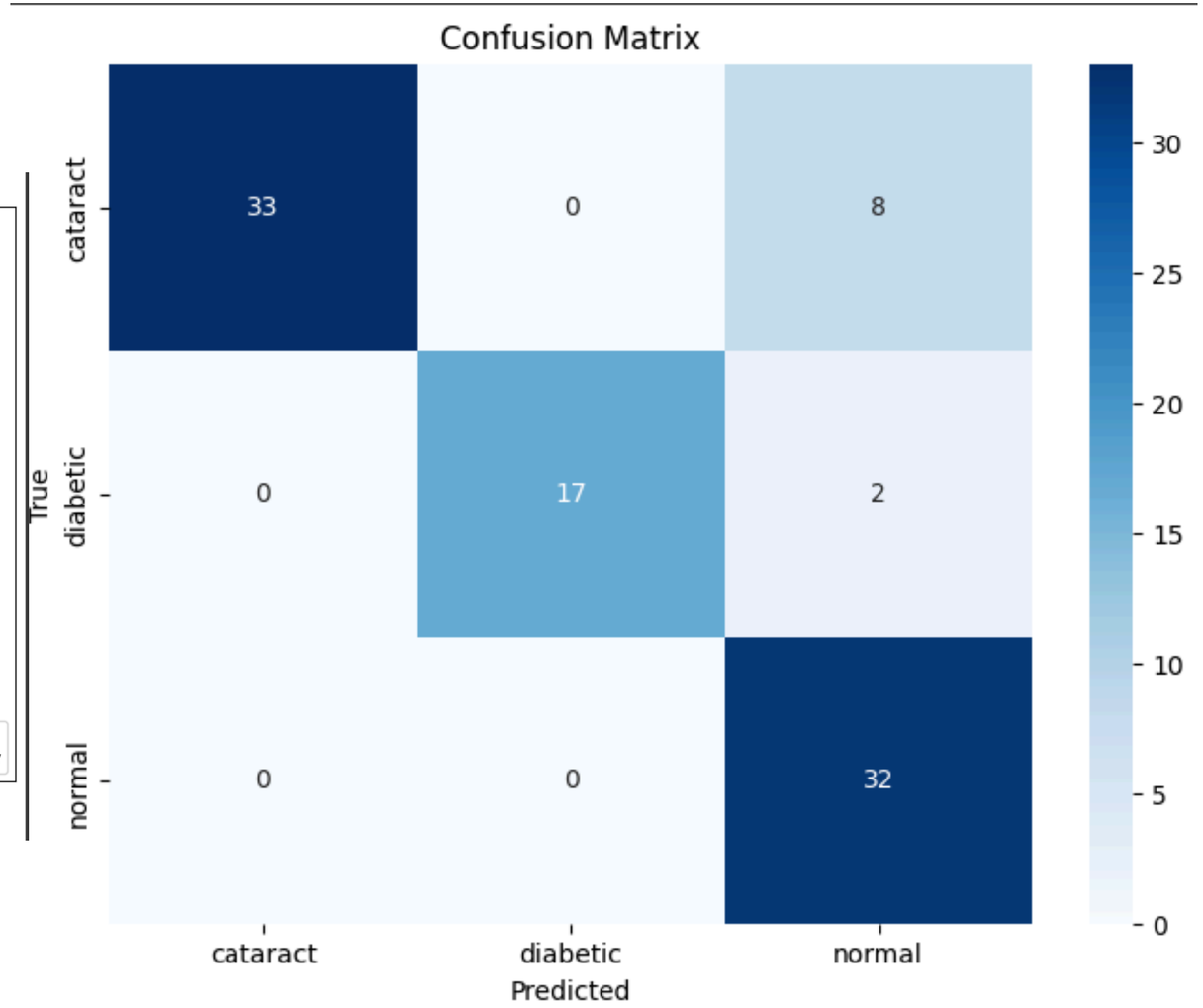
MODEL EVALUATION ON OFFLINE MODE

Class Name	Precision	Recall	F1 Score
diabetic-retinopathy	1	0.89	0.94
normal	0.76	1	0.86
cataract	1	0.80	0.89
Accuracy			0.89

MODEL EVALUATION ON OFFLINE MODE







Flask API Integration





Setup

Configure HTTP API Endpoint and Import Library



Image Handling

Manage image uploads and process API requests

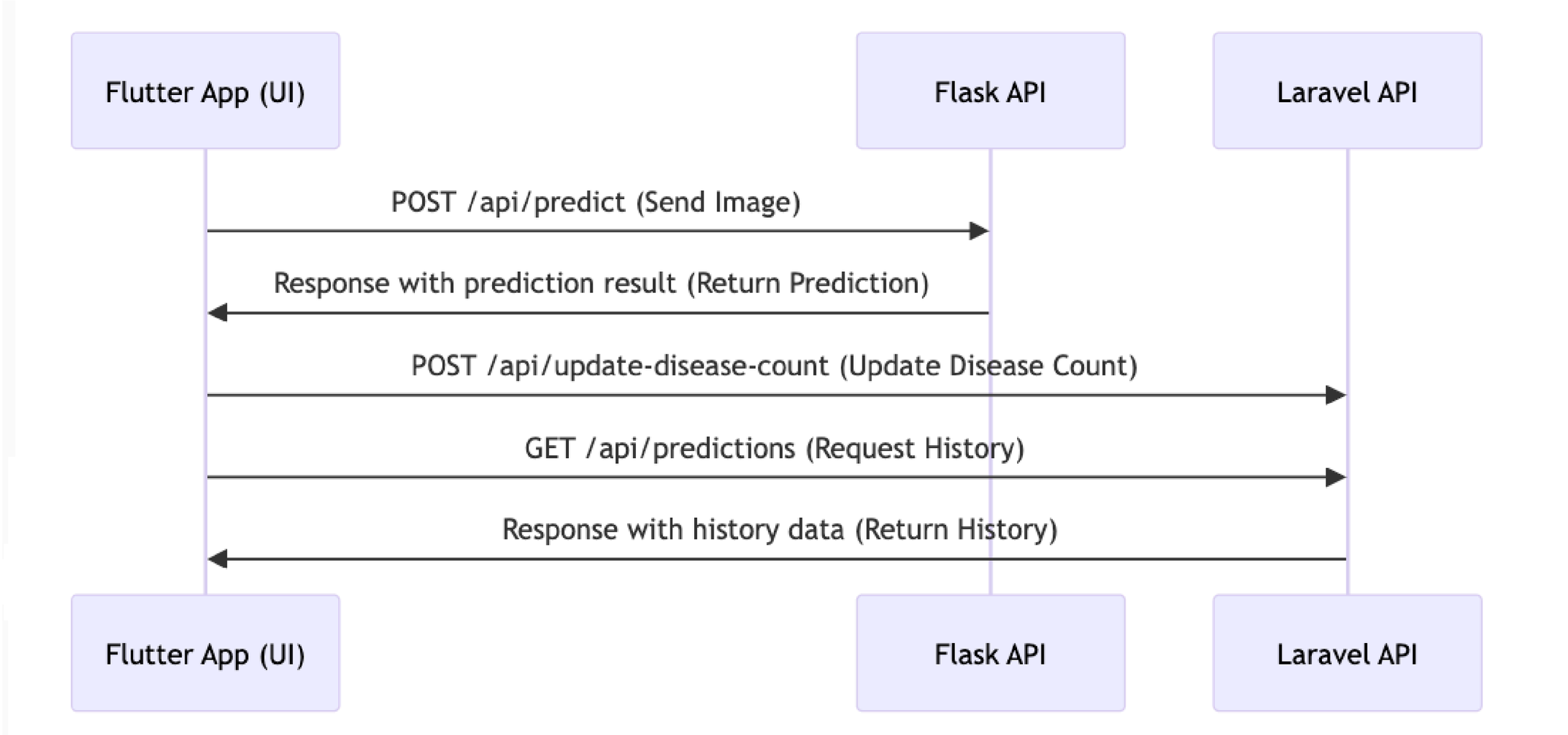


Display Results

Return results from the Flask API

MOBILE APPLICATION

Disease detection



5

CONCLUSION

RESULT



Mobile Application

100% COMPLETED



Web Application

90% COMPLETED

Overall



EXPERIENCES

- Work as team
- Increased responsibility with work
- Understand more about PyTorch, Flask API
- Getting better of research work with VGG16, ResNet50, and DenseNet121 model
- Understand the metric of model evaluation



CHALLENGES



Time Constraint



New to Deep Leaning



Misunderstood requirement

FUTURE WORK



Goal #1

Enhance Security and Improve UI



Goal #2

Get the real data from the Khmer hospitals



Goal #3

Deploy our application

DEMO

REFERENCES

[1] Dataset for different eye diseases. (n.d.). Kaggle: Your Machine Learning and Data Science Community.

[2] Eye_diseases_classification. (n.d.). Kaggle: Your Machine Learning and Data Science Community.

[3] A global review of publicly available datasets for ophthalmological imaging: barriers to access, usability, and generalisability. (2020, October 1). The Lancet Digital Health.

[4] VGG-16 | CNN model. (2024, March 21). GeeksforGeeks.

[5] Persson, A. (2023, April 15). Machine Learning Collection. GitHub.

[6] Nabi, J. (2019, March 16). Hyper-parameter Tuning Techniques in Deep Learning. Medium.

[7] Kundu, N. (2023, January 23). Exploring ResNet50: An In-Depth Look at the Model Architecture and Code Implementation. Medium.

[8] Image Classification Using Pytorch and Convolutional Neural Network. (n.d.). Www.youtube.com.